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Acceptability of Academic Large Language Model for Patients Seeking Health Information

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Abstract

The use of large language models (LLMs) in healthcare poses challenges related to data security, accuracy, bias, and usability, which can hinder their effectiveness in enhancing patients' ability to locate trusted health information. We conducted a pilot study of a local LLM named "SAM," built upon the Llama 7B architecture. Participants engaged with SAM by submitting health-related questions across five health domains. Following their interactions, participants completed the System Usability Scale (SUS) to evaluate the usability of the model. Of the ten participants, eight (80%) were female, and two (20%) were male. The highest-rated theme was ease of learning, with participants strongly agreeing that most people would quickly learn to use the chatbot (Mean = 4.7, SD = 0.46). Developing a local LLM for patient health information must tackle healthcare barriers. By enhancing data security, personalizing responses, and increasing user familiarity, SAM can improve patient engagement and outcomes.

Keywords

Generative AI; Usability Evaluation; User Experience

1. Introduction

The integration of large language models (LLMs) in healthcare presents several significant challenges from a user experience perspective. One of the primary issues is data security and privacy. LLMs need large amounts of data, often containing sensitive patient information. Protecting this data's confidentiality and integrity is crucial, but breaches and misuse are still concerns [1]. Accuracy and reliability are essential. LLMs may generate incorrect or misleading information, which can have serious consequences [2]. Bias and fairness are also problematic, as these models can inadvertently perpetuate existing biases present in the training data [3].

The impact of these challenges is profound. Inaccurate information can result in misdiagnoses or inappropriate treatment recommendations, compromising patient safety and care quality [2]. Bias in LLMs can exacerbate health disparities, affecting the fairness and

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equity of healthcare delivery [3]. Usability challenges can impede the adoption of LLMs, as end-users may find it difficult to integrate these tools effectively into their workflows. Collectively, these issues diminish the potential benefits of LLMs in improving healthcare delivery and patient outcomes.

A promising solution to these challenges is to create a local LLM specifically designed for healthcare applications. Organizations can improve data security and privacy by developing and deploying LLMs within local healthcare settings while ensuring compliance with regulations like HIPAA [4]. Local instances can be fine-tuned with domain-specific data, improving accuracy and reducing bias [5]. Moreover, end-users can customize the user interface and functionality to better meet their needs. This customization enhances usability and increases acceptability, improving healthcare delivery and patient outcomes. This study examined the acceptability of a home-grown LLM by potential end-users seeking health information.

2. Methods

A pilot study was conducted at a Southeastern medical center in the US. We developed a local LLM named “SAM” to facilitate patient communication in the context of healthcare. To build SAM, we adopted the Llama 7B model, an open-source LLM developed by Meta. The Llama 7B model is designed to handle a wide range of natural language processing tasks and is particularly suited for healthcare-related inquiries, thanks to its robust language capabilities and ability to process complex, context-sensitive health information.

The model was deployed on a secure server at the institution, ensuring all data processing remained within the institution’s infrastructure. The server setup prioritized security, employing multi-layered access controls and encryption mechanisms to safeguard patient information. All communication between the client-side interface and the model was encrypted using TLS protocols to ensure data privacy during interactions.

We developed a custom-built user interface (UI) to interact with the LLM. The UI was designed to facilitate seamless communication between users and SAM. It followed a structure similar to widely used models like ChatGPT, where a log of questions and responses is presented to the user on the screen. This allowed users to track their interactions with SAM easily.

Each participant was assigned a unique user ID, which allowed SAM to recognize and log their sessions. This unique ID enabled the model to track and learn about each user’s health preferences, concerns, and past interactions. Consequently, SAM could personalize its responses based on the user’s history and characteristics, tailoring its replies to align with the user’s specific context and needs.

2.1. Participant Interaction

Participants were asked to engage with SAM by submitting at least five health-related questions. These questions were required to fall within one of the five domains: Primary Care, Nutrition, Lifestyle, Cancer, and Pediatrics. After interacting with SAM, participants

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were prompted to complete the System Usability Scale (SUS) electronically through a tablet. The SUS is a widely used tool to assess the usability of systems, providing a quick and reliable measure of user satisfaction. It consists of 10 statements, each rated on a 5-point Likert scale, ranging from strongly agree to strongly disagree. The scores from the SUS were used to calculate an overall usability score, with a higher score indicating better perceived usability of the system.

In addition to completing the SUS, participants were also asked to list the most favorable features of SAM and areas where they felt improvements could be made. This feedback was crucial for understanding the strengths and weaknesses of SAM from the user's perspective and for identifying potential improvements in both the user interface and the model's functionality.

The SUS is a widely recognized tool for evaluating the usability of systems and provides an overall score that reflects users' perceptions of a system's ease of use. The scale consists of 10 statements that assess various aspects of the system, such as its learnability, efficiency, and user satisfaction. Participants were asked to rate each statement on a 5-point Likert scale, with responses ranging from strongly agree to strongly disagree. The total SUS score is then calculated by adding the scores for each statement, resulting in a score between 0 and 100. Higher scores indicate better usability.

For the statistical analysis of the study, we employed both descriptive and inferential techniques to assess the usability and feedback from participants. Descriptive statistics were used to summarize the demographic characteristics of the participants and their interactions with SAM, including the mean and standard deviation (SD). The System Usability Scale (SUS) scores were analyzed using the mean and SD to assess overall usability, with higher scores indicating better user satisfaction with the system. In addition, a word cloud was generated to analyze the emerging themes from the open-ended questions. This word cloud visualized the most frequently mentioned terms and phrases, offering insights into the participants' most favorable features of SAM and areas where improvements could be made.

3. Results

Of the ten participants, eight (80%) were female, and two (20%) were male. The most common age group was 36–40 years (4, 40%), followed by 31–35 years (2, 20%), 41–45 years (2, 20%), and 22–30 years (2, 20%). Regarding AI experience, seven (70%) participants reported using AI tools for 2–3 years, while three (30%) had 1 year of experience. In terms of estimated usage of large language models (LLMs) per week, six (60%) participants reported 0–10 hours, two (20%) reported 11–20 hours, and one (10%) reported 21–30 hours, Table 1.

The highest-rated theme was ease of learning, with participants strongly agreeing that most people would quickly learn to use the chatbot (Mean = 4.7, SD = 0.46). Similarly, user confidence was highly rated, as participants felt very confident using the chatbot (Mean = 4.7, SD = 0.64). These findings suggest that SAM is perceived as intuitive and user-friendly. On the other hand, the lowest-rated theme was perceived complexity, with participants

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strongly disagreeing that the chatbot was unnecessarily complex (Mean = 1.1, SD = 0.30). Additionally, perceived cumbersome use was also rated low (Mean = 1.4, SD = 0.92), further reinforcing that users did not find the chatbot difficult or burdensome to interact with. Overall, the results indicate that SAM is viewed as an accessible and easy-to-use tool, with minimal perceived complexity or barriers to adoption.

Participants highlighted several positive aspects of SAM, particularly its ease of use, with many describing the interface as simple and intuitive. Personalization was another key theme, as users appreciated tailored responses to their specific needs. The clarity of the conversation flow, especially the format that displayed questions and answers clearly, was valued. Participants also noted SAM's efficiency in providing quick and relevant answers, along with its safety and reliability, noting that it encourages seeking professional medical advice when necessary. Overall, SAM was seen as a supportive and responsible tool.

Participants highlighted areas for improvement in SAM, primarily focusing on interface design and usability. Suggestions included a more visually appealing layout with bold titles and bullet points for better readability. Users found the submission process cumbersome, preferring a submit button or an indicator for processing queries. Response time was also a concern, as some participants noted that SAM was slower than other models. Additionally, users wanted more in-depth information, such as detailed explanations and references to scientific guidelines. While some users had no complaints, these themes point to opportunities for enhancing SAM's user experience.

4. Discussion

The findings of our pilot study highlight the positive reception of participants, which suggests that when tailored to healthcare contexts, LLMs can significantly enhance patient interaction. The acceptability of SAM highlights the importance of user-centered design in healthcare AI tools. Its customized user interface simplified interactions and empowered users, fostering a sense of control and personalization essential for effective healthcare relationships. This supports literature emphasizing usability as a key factor in technology adoption in clinical settings.

One of the primary advantages of deploying SAM within the confines of our institution is the enhanced data security and privacy it affords. Participants expressed positive sentiments regarding the assurance that their sensitive health information would remain secure. Given the increasing concerns surrounding data breaches in healthcare, our approach underscores the importance of embedding data protection in designing and deploying technological solutions in this domain.

Our findings suggest that tools like SAM could improve communication, empower patient decisions, and enhance healthcare quality. To achieve this, collaboration among AI developers, healthcare professionals, and policymakers is essential to ensure ethical deployment, promote equity, and enhance patient safety.

Despite these promising results, our study has some limitations that must be acknowledged. First, the sample size was relatively small and comprised participants from a single medical

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center, which may limit the generalizability of our findings. A larger, more diverse cohort spanning multiple healthcare settings would help validate the effectiveness and acceptability of local LLMs like SAM.

In conclusion, while the pilot study demonstrates an encouraging acceptance of the local LLM among participants, future research should focus on expanding the user base, addressing ongoing concerns related to bias, accuracy, and usability, and assessing the long-term impact of LLM integration in healthcare settings. By advancing our understanding of these dynamics, we can better harness the power of generative AI to improve patient communication and health outcomes.

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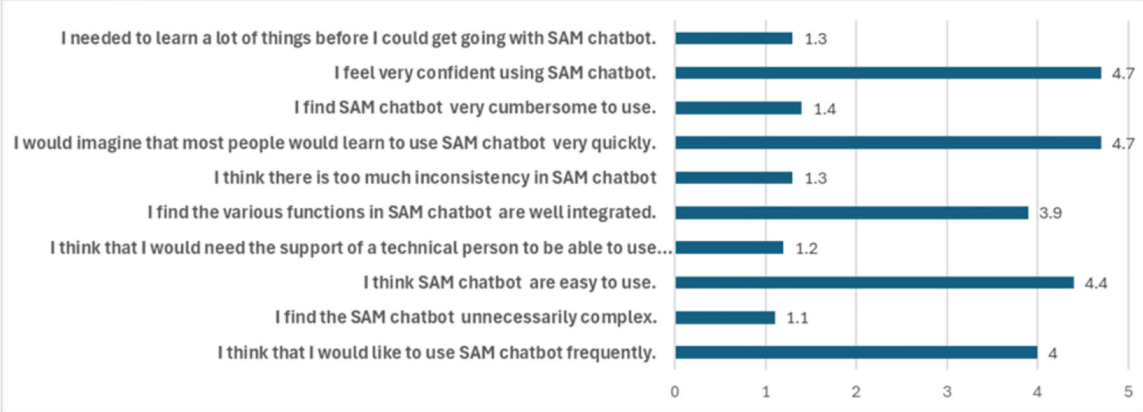


Figure 1.
Results from the System Usability Scale (SUS) survey.

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Figure 2.
Word Cloud of the most favorable (A) and unfavorable (B) features of SAM.

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Table 1.

Study participant characteristics

Characteristic	Frequency (n)	Percentage (%)
Gender		
Female	8	80
Male	2	20
Age Group (years)		
22–30	2	20
31–35	2	20
36–40	4	40
41–45	2	20
Years Using AI Tools		
1 year	3	30
2–3 years	7	70
Estimated LLM Usage (hours/week)		
0–10	6	60
11–20	2	20
21–30	1	10